

Question **1**

Complete

Mark 0.00 out of 1.00

All the following statements correctly describe ketone bodies, EXCEPT:

- ☐ a. They are utilized by the liver during long-term starvation.
- ☐ b. Their precursors are Acetyl CoA
- ☒ c. They may be synthesized during starvation.
- ☐ d. They may be synthesized in uncontrolled Diabetes Mellitus.
- ☐ e. They are utilized by the cardiac muscles.

Question **2**

Complete

Mark 1.00 out of 1.00

Catabolism of histidine might be impaired due to the dietary deficiency of

- ☒ a. Folic acid
- ☐ b. Vitamin B1 Thiamin)
- ☐ c. Vitamin B3 (Niacin)
- ☐ d. Biotin
- ☐ e. Vitamin B6 (pyridoxal phosphate)

Question **3**

Complete

Mark 1.00 out of 1.00

Which one of the following would be expected to occur after acute alcohol ingestion?

- ☒ a. Lactic acidosis
- ☐ b. An increase in gluconeogenesis
- ☐ c. Activation of fatty acid oxidation
- ☐ d. An increase in the NAD⁺/NADH ratio
- ☐ e. Increase pyruvate concentration

Question 4

Complete

Mark 0.00 out of 1.00

Which of the following enzymes controls substrate availability for β -oxidation pathway?

- ☐ a. Acyl CoA Dehydrogenase
- ☐ b. Hormone sensitive lipase.
- ☐ c. Lipoprotein lipase.
- ☐ d. Acyl CoA-Carnitine transferase.
- ☒ e. Acyl CoA carboxylase.

Question 5

Complete

Mark 0.00 out of 1.00

"Cytochrome P450 reductase" which catalyzes hydroxylation of drugs requires the coenzyme:

- ☐ a. FAD
- ☐ b. NAD⁺
- ☐ c. NADPH + H⁺
- ☒ d. NADP⁺
- ☐ e. NADH + H⁺

Question 6

Complete

Mark 0.00 out of 1.00

The deamination process that converts cytosine into uracil is an example of

- ☒ a. Non-oxidative deamination
- ☐ b. Oxidative deamination
- ☐ c. Transdeamination
- ☐ d. Hydrolytic deamination
- ☐ e. Deamidation

Question **7**

Complete

Mark 0.00 out of 1.00

Which of the following tissues cannot utilize ketone bodies?

- ☐ a. Brain.
- ☐ b. Kidneys.
- ☐ c. Liver.
- ☒ d. Adipose tissue
- ☐ e. Skeletal muscles.

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following does not occur due to deficiency of enzymes for urea cycle?

- ☐ a. Mental retardation
- ☐ b. Hyperammonemia
- ☒ c. Uremia
- ☐ d. Vomiting
- ☐ e. Convulsions

Question **9**

Complete

Mark 0.00 out of 1.00

The key enzyme in the regulation of fatty acid synthesis is

- ☐ a. Acetyl CoA carboxylase
- ☐ b. Citrate Synthase and Lyase
- ☐ c. Carnitine acyl transferase
- ☐ d. Fatty acyl CoA carboxylase
- ☒ e. Fatty acyl synthetase

Question **10**

Complete

Mark 0.00 out of 1.00

Which of the following is NOT synthesized from cholesterol?

- ☐ a. Corticosteroid
- ☒ b. Vitamin D3
- ☐ c. Ketone bodies
- ☐ d. Bile acids
- ☐ e. Steroids

Question **11**

Complete

Mark 0.00 out of 1.00

Among the following protein digestive enzymes, which one originates from duodenal mucosa?

- ☐ a. chymotrypsin
- ☐ b. Enteropeptidase
- ☐ c. Carboxypeptidases
- ☒ d. Elastases
- ☐ e. Trypsin

Question **12**

Complete

Mark 0.00 out of 1.00

What is the molecular basis for the statement "fats burn in the flame of carbohydrates"?

- ☐ a. Fatty acids and glucose oxidized simultaneously.
- ☐ b. Fatty acids and glucose are oxidized in different time but the glucose is used to synthesis fats
- ☐ c. Acetyl CoA is oxidized completely in the presence of oxaloacetate in TCA cycle.
- ☐ d. Acetyl CoA is the common product of fatty acid and glucose oxidation
- ☒ e. Fats hydrolyzed in the presence of carbohydrates.

Question **13**

Complete

Mark 0.00 out of 1.00

The main cytosolic catabolism of threonine results in the generation of

- ☐ a. Pyruvate
- ☐ b. Citrate
- ☐ c. Succinyl CoA
- ☐ d. A-ketoglutarate
- ☒ e. Acetyl CoA

Question **14**

Complete

Mark 0.00 out of 1.00

The number of LDL receptors is down-regulated by:

- ☐ a. Decreased intracellular cholesterol
- ☒ b. Higher levels of estrogens.
- ☐ c. Increased intracellular cholesterol.
- ☐ d. Higher ratio of unsaturated fatty acids in diet.
- ☐ e. Increased LDL concentration.

Question **15**

Complete

Mark 0.00 out of 1.00

In an individual exposed to xenobiotics, such as alcohol, medications, and other foreign chemicals, the risk for free-radical injury increases through one of the following mechanisms?

- ☐ a. Induction of oxidases in peroxisomes
- ☐ b. Induction of enzymes containing cytochrome P450
- ☒ c. Production of ionizing radiation
- ☐ d. Reaction of O₂ with Coenzyme-Q
- ☐ e. Production of hydrogen peroxide in the manufacturing process, such that hydrogen peroxide is present in the ingested materials

Question **16**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the drug interactions between phenobarbital and warfarin?

- ☐ a. Phenobarbital represses CYP2C9, and this results in increased catabolism of warfarin.
- ☐ b. Warfarin induces CYP2C9, and this results in decreased catabolism of phenobarbital.
- ☐ c. Phenobarbital induces CYP2C9, and this results in decreased catabolism of warfarin.
- ☒ d. Phenobarbital induces CYP2C9, and this results in increased catabolism of warfarin.
- ☐ e. Warfarin induces CYP2C9, and this results in increased catabolism of phenobarbital.

Question **17**

Complete

Mark 0.00 out of 1.00

Homocystinuria is caused by defective process of

- ☐ a. Valine degradation
- ☐ b. Leucine degradation
- ☐ c. conversion of phenyl alanine to tyrosine
- ☒ d. conversion of propionyl CoA to succinyl CoA
- ☐ e. Degradation of Methionine

Question **18**

Complete

Mark 0.00 out of 1.00

HMG-CoA reductase activity is increased in the presence of:

- ☐ a. Epinephrine
- ☒ b. Lovastatin
- ☐ c. Insulin
- ☐ d. Endogenous cholesterol levels
- ☐ e. Glucagon

Question **19**

Complete

Mark 0.00 out of 1.00

Which of the following enzymes controls substrate availability to β -oxidation pathway?

- ☐ a. Acyl CoA carboxylase.
- ☐ b. Lipoprotein lipase.
- ☒ c. Hormone sensitive lipase. Ping
- ☐ d. Acyl CoA Dehydrogenase
- ☐ e. Acyl CoA-Carnitine transferase.

Question **20**

Complete

Mark 0.00 out of 1.00

Where is lipoprotein lipase present?

- ☐ a. In the inner mitochondrial membrane of all extrahepatic tissues.
- ☒ b. In the cytosol of adipose tissue.
- ☐ c. On the hepatocytes, attached with the LDL-receptor.
- ☐ d. In the liver
- ☐ e. On the endothelial cells of the capillaries in cardiac and skeletal muscles.

Question **21**

Complete

Mark 0.00 out of 1.00

Supplement of additional L-arginine could be a solution for an individual deficient in:

- ☐ a. Ornithine transcarbamoylase
- ☒ b. Carbamoyl phosphate synthase
- ☐ c. Argininosuccinase
- ☐ d. Arginase
- ☐ e. Argininosuccinate synthase

Question **22**

Complete

Mark 0.00 out of 1.00

Which of the following is not used for fatty acid synthesis?

- ☐ a. Biotin.
- ☐ b. Cobalamin
- ☐ c. NADPH
- ☒ d. Bicarbonate.
- ☐ e. Acetyl CoA

Question **23**

Complete

Mark 0.00 out of 1.00

One of the following lipoproteins is used to transport endogenously synthesized Triacylglycerols from the liver to peripheral tissues

- ☐ a. HDL
- ☐ b. LDL
- ☐ c. VLDL
- ☐ d. Chylomicron
- ☒ e. Chyme

Question **24**

Complete

Mark 0.00 out of 1.00

How many ATPs are formed during complete oxidation of Stearic acid (18:0)?

- ☐ a. 122
- ☐ b. 154
- ☒ c. 106
- ☐ d. 148
- ☐ e. 139

Question **25**

Complete

Mark 1.00 out of 1.00

Which of the following pairs shows a mismatch between the source and product of amino acid metabolism

- ☒ a. Tryptophan T Nitric oxide
- ☐ b. Tyrosine T Melanin
- ☐ c. Glutamate G γ -amino butyric acid
- ☐ d. Serine S Glycine
- ☐ e. Histidine H Histamine

Question **26**

Complete

Mark 1.00 out of 1.00

After digestion and absorption of dietary proteins, one of the following amino acid is preferred to be used by enterocytes as source of energy.

- ☐ a. Asparagine
- ☐ b. Glycine
- ☒ c. Glutamate
- ☐ d. Arginine
- ☐ e. Proline

Question **27**

Complete

Mark 0.00 out of 1.00

Which of the following is TRUE about fatty acid synthesis?

- ☐ a. Acetyl-CoA carboxylase controls the pace of fatty acid synthesis.
- ☐ b. The two units of the dimeric fatty acid synthetase are placed in Head-to-Head manner.
- ☐ c. HMG-CoA reductase is the rate limiting step.
- ☐ d. Fatty acids synthase is monomeric enzyme
- ☒ e. One malonyl-CoA and seven Acetyl-CoA are utilized for palmitic acid synthesis.

Question **28**

Complete

Mark 0.00 out of 1.00

The enzyme for protein degradation that is commonly activated by a protein originated from Human Papiloma virus in cervical carcinoma patients is

- ☐ a. Enteropeptidase
- ☐ b. Ubiquitin-Activating Enzyme
- ☒ c. Exopeptidase
- ☐ d. Carboxypeptidase
- ☐ e. Ubiquitin-Protein Ligase

Question **29**

Complete

Mark 0.00 out of 1.00

Which of the following tissues lacks Glycerol Kinase activity?

- ☐ a. Kidneys.
- ☒ b. All of the above
- ☐ c. Adipose tissue.
- ☐ d. Small intestine.
- ☐ e. Liver.

Question **30**

Complete

Mark 0.00 out of 1.00

Co-lipase increases the activity of pancreatic lipase by:

- ☐ a. Co-lipase inhibits the activity of pancreatic lipase.
- ☐ b. Protecting the enzyme from the inhibitory effects of bile salts.
- ☐ c. Forming pancreatic lipase-Colipase complex
- ☒ d. By cleaving some peptides of pancreatic lipase
- ☐ e. Helping pancreatic lipase to anchor dietary lipids.

Question **31**

Complete

Mark 0.00 out of 1.00

Induction of CYP2E1 would be expected to result in which of the following?

- ☐ a. protection from alcohol-induced hepatic necrosis
- ☐ b. decreased rate of ethanol clearance from the blood
- ☒ c. decreased rate of acetaldehyde production
- ☐ d. reduced likelihood of free radical production
- ☐ e. increased level of alcohol tolerance

Question **32**

Complete

Mark 1.00 out of 1.00

During heavy exercise, apart from carbohydrate and fatty acids, the muscle also preferably catabolize one of the following amino acid as alternative source of energy

- ☒ a. Valine
- ☐ b. Alanine
- ☐ c. Tyrosine
- ☐ d. Threonine
- ☐ e. Serine

Question **33**

Complete

Mark 0.00 out of 1.00

During fasting one of these events does not take place:

- ☐ a. Hormone- sensitive lipase is activated.
- ☐ b. Glucose uptake by adipose tissue is increased.
- ☐ c. There will be an increased synthesis of glucose from non-carbohydrate source
- ☒ d. Plasma non-esterified fatty acids (NEFAs) are increased.
- ☐ e. Triacylglycerols are mobilized from adipose tissue.

Question **34**

Complete

Mark 0.00 out of 1.00

Which of the following condition promotes/ increases urea formation?

- ☐ a. Activation of carbamoyl phosphate synthase II
- ☐ b. Decrease in ornithine synthesis
- ☐ c. Increase in N-acetyl glutamate
- ☒ d. Decrease in Glutamate
- ☐ e. Increase in Urea concentration

Question **35**

Complete

Mark 0.00 out of 1.00

Glycine can be catabolized by D-amino acid oxidase because

- ☐ a. It has no asymmetric carbon
- ☒ b. It can polymerize with other amino acids
- ☐ c. It is the smallest in size
- ☐ d. It is polar and easy to transport
- ☐ e. It is the most abundant

Question **36**

Complete

Mark 0.00 out of 1.00

Which of the following is the primary ketone body?

- ☐ a. HMG-CoA
- ☐ b. Acetoacetate
- ☒ c. Acetone.
- ☐ d. Hydroxyl methyl glutarate.
- ☐ e. Betahydroxy butyrate.

Question **37**

Complete

Mark 1.00 out of 1.00

Which of the following is true about phenylketonuria (PKU)

- ☐ a. PKU results in accumulation of tyrosine
- ☐ b. PKU results in reduced synthesis of phenyl pyruvate
- ☐ c. PKU results in increase synthesis of homogentisate
- ☐ d. PKU results in deficiency of tyrosine aminotransferase
- ☒ e. PKU results in inhibition of pyruvate carboxylase

Question **38**

Complete

Mark 0.00 out of 1.00

Which one of the following enzymes is used for activation of fatty acids in intestine?

- ☐ a. Enoyl CoA hydratase
- ☐ b. Acyl transferase
- ☐ c. Fatty acyl CoA dehydrogenase
- ☐ d. Fatty acyl CoA synthetase.
- ☒ e. Enoyl CoA dehydrogenase.

Question **39**

Complete

Mark 1.00 out of 1.00

A researcher has generated a cell line in which the γ -glutamyl cycle is defective, and glutathione cannot be synthesized. Which radical species might you initially expect to accumulate in this cell?

- ☐ a. Superoxide
- ☐ b. Nitrous oxide
- ☐ c. Nitrogen dioxide
- ☒ d. Hydrogen peroxide
- ☐ e. Peroxynitrate

Question **40**

Complete

Mark 0.00 out of 1.00

Which subfamily of cytochrome P-450 enzyme is responsible for the highest fraction of clinically important drug interactions resulting from metabolism of:

- ☒ a. CYP2D6
- ☐ b. CYP3A4
- ☐ c. CYP2A6
- ☐ d. CYP2E1
- ☐ e. CYP1A2

Question **41**

Complete

Mark 1.00 out of 1.00

Kidney disease possibly related with the following biochemical processes Except

- ☐ a. Catalysis by D-amino acid oxidase
- ☐ b. Catabolism of glycine by kidney
- ☐ c. Production of calcium oxalate
- ☒ d. Increased activity of glutaminase
- ☐ e. Increased intake of oxalate-rich diet

Question **42**

Complete

Mark 0.00 out of 1.00

The acetyl-CoA for fatty synthesis acid comes from all, EXCEPT

- ☐ a. Propionyl CoA catabolism
- ☒ b. Glucose catabolism.
- ☐ c. β -oxidation of fatty acids.
- ☐ d. Ketogenic amino acids
- ☐ e. Pyruvate catabolism

Question **43**

Complete

Mark 0.00 out of 1.00

Which one of the following statements is Correct about the absorption of lipids from the intestine?

- ☐ a. Release of fatty acids from triacylglycerol in the intestine is inhibited by bile salts.
- ☐ b. Dietary triacylglycerol is partially hydrolyzed and absorbed as free fatty acids and monoacylglycerol.
- ☒ c. Dietary triacylglycerol must be hydrolyzed to free fatty acids and glycerol before absorption.
- ☐ d. End products of lipid digestion are actively absorbed with sodium
- ☐ e. Fatty acids that contain ten carbons or less are absorbed and enter the circulation primarily via the lymphatic system.

Question **44**

Complete

Mark 0.00 out of 1.00

Which of the following is not true about glutaminase

- ☐ a. Catalyze transamination reaction
- ☐ b. Abundant in liver
- ☐ c. A mitochondrial enzyme
- ☒ d. Important in urea synthesis
- ☐ e. Produce glutamate

Question **45**

Complete

Mark 1.00 out of 1.00

The following biochemical process is associated with serine catabolism.

- ☒ a. Non-oxidative deamination
- ☐ b. Dehydrogenation involving NAD⁺
- ☐ c. Reduction of FAD
- ☐ d. Transamination involving T-ketoglutarate
- ☐ e. Generation of peroxide

Question **46**

Complete

Mark 0.00 out of 1.00

Which of the following is not true about kynurenine pathway?

- ☐ a. Tryptophan indole ring produce acetoacetate
- ☐ b. Tryptophan catabolism involves di- and monooxygenases
- ☐ c. The major pathway of tryptophan metabolism
- ☐ d. Tryptophan produces serotonin
- ☒ e. Tryptophan side chain produces alanine

Question **47**

Complete

Mark 0.00 out of 1.00

The form in which most dietary lipids are packaged and exported from the intestinal mucosal cells to lymph is as:

- ☐ a. Chylomicron.
- ☐ b. VLDL.
- ☒ c. Micelle
- ☐ d. Free fatty acids.
- ☐ e. LDL.

Question **48**

Complete

Mark 0.00 out of 1.00

The drug "isoniazid" (isonicotinic acid hydrazide) used in the treatment of tuberculosis is detoxicated by:

- ☐ a. Cytochrome P-450 enzyme species
- ☐ b. Acetylation
- ☐ c. Glutathione
- ☒ d. Methylation
- ☐ e. 'Active' sulphate

Question **49**

Complete

Mark 0.00 out of 1.00

Which of the following metabolic condition do not lead to fatty acid synthesis?

- ☐ a. Presence of high amino acids
- ☐ b. Presence of low carbohydrates
- ☐ c. Presence of low fat in the body
- ☒ d. Presence of low protein
- ☐ e. Presence of high sugars

Question **50**

Complete

Mark 0.00 out of 1.00

Triacylglycerol molecules stored in adipose tissue represent the major reserve of substrate providing energy during a prolonged fast. During such a fast:

- ☒ a. Free fatty acids are produced at a high rate in the plasma by the action of lipoprotein lipase on chylomicrons.
- ☐ b. Hormone- sensitive lipase is phosphorylated by a cyclic AMP-activated protein kinase
- ☐ c. Glycerol produced by the degradation of triacylglycerol is an energy sources for adipocytes and fibroblasts.
- ☐ d. The free fatty acids in the plasma transported by glucuronic acid
- ☐ e. The stored fatty acids are released from adipose tissue into the plasma as components of the serum lipoprotein particle, VLDL.

Question **51**

Complete

Mark 0.00 out of 1.00

Which of the following participates in the reverse transport of cholesterol:

- ☐ a. VLDL.
- ☐ b. sLDL
- ☒ c. LDL.
- ☐ d. Chylomicrons.
- ☐ e. HDL.

Question **52**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the effects of polymorphisms of CYP2A6?

- ☐ a. People with the inactive (null) allele are more likely to become tobacco-dependent smokers because cytochrome inactivates nicotine to cotinine.
- ☐ b. People with the inactive (null) allele are less likely to become tobacco-dependent smokers because cytochrome activates nicotine to cotinine.
- ☒ c. People with the inactive (null) allele are less likely to become tobacco-dependent smokers because this cytochrome inactivates nicotine to cotinine.
- ☐ d. People with the inactive (null) allele are more likely to become tobacco-dependent smokers because cytochrome activates nicotine to cotinine.
- ☐ e. People with the active allele are less likely to become tobacco-dependent smokers because cytochrome inactivates nicotine to cotinine.

Question **53**

Complete

Mark 1.00 out of 1.00

The endogenous protein that is catabolized through proteasomal pathway include

- ☐ a. Long-lived intracellular proteins
- ☒ b. Short-living intracellular proteins
- ☐ c. Membrane-associated proteins
- ☐ d. Turnover of organelles
- ☐ e. Extracellular proteins

Question **54**

Complete

Mark 0.00 out of 1.00

Which of the following enzymes is key for the utilization of ketone bodies:

- ☐ a. Thiolase
- ☒ b. Thioesterase
- ☐ c. Thiophorase
- ☐ d. Thiokinase
- ☐ e. HMG-CoA synthase

Question **55**

Complete

Mark 0.00 out of 1.00

During absorption of dietary amino acid, the most common amino acids catabolized by enterocytes are

- ☐ a. Leucine and lysine
- ☐ b. Glutamine and threonine
- ☐ c. Valine and isoleucine
- ☒ d. Glycine and alanine
- ☐ e. Asparagine and glutamine